

Project Proposal — Artificial Personal Digital Twin

1. Project Title

Artificial Personal Digital Twin: A Multi-Agent System for Intelligent Life Management

2. Objective

The objective of this project is to design and develop an **Artificial Personal Digital Twin**, a multi-agent AI system that represents the user digitally and assists in:

- Health monitoring
- Fitness and nutrition planning
- Financial decision-making
- Daily productivity optimization

The system will act as a **virtual version of the user**, capable of analyzing data, making decisions, and recommending or executing actions.

3. Problem Statement

Today's digital tools (health apps, banking apps, calendars, fitness platforms) have major limitations:

- Fragmented systems (each app works alone)
- No communication between domains
- Lack of personalization
- Limited intelligence — cannot prioritize or act holistically

Users often have **data overload but no actionable insights**.

General User Problems

Modern users face multiple challenges that current systems fail to solve:

- a. Users **can't optimize their time**: balancing work, studies, health, and social life is a constant struggle.
- b. Important **personal data gets lost**: medical reports, finances, fitness progress — scattered across apps.
- c. Generic health or fitness plans **don't work**: your body, goals, and lifestyle are unique, yet advice is one-size-fits-all.
- d. Money management is confusing: users often ask, *"Where did my money go?"* but apps don't explain it clearly.
- e. AI can draft emails or schedule meetings — **but never in your tone or style**, so you can't fully trust it.

Challenges for International Students

- New countries = new systems, new lifestyle, new challenges.
- Hard to manage **budget, time, and health** simultaneously.
- Generic advice doesn't adapt to cultural differences or personal routines.

People don't need more apps.
They need a **single intelligent system** that understands them, connects all life areas, and **acts like a personal assistant / digital twin**.

4. Proposed Solution

We propose an **Artificial Personal Digital Twin**, powered by a **multi-agent architecture**, where each agent models a specific aspect of the user.

The system continuously:

- Observes user data
- Learns behavior
- Simulates decisions
- Suggests or executes actions

5. System Components (Agents)

Agent	Role
Health Agent	Analyze medical reports & detect trends
Fitness Agent	Generate workouts & nutrition plans
Finance Agent	Track spending & predict financial status
Twin Core Agent	Simulates user behavior & prioritizes decisions
Action Agent	Executes tasks (reminders, scheduling, alerts)

6. System Architecture

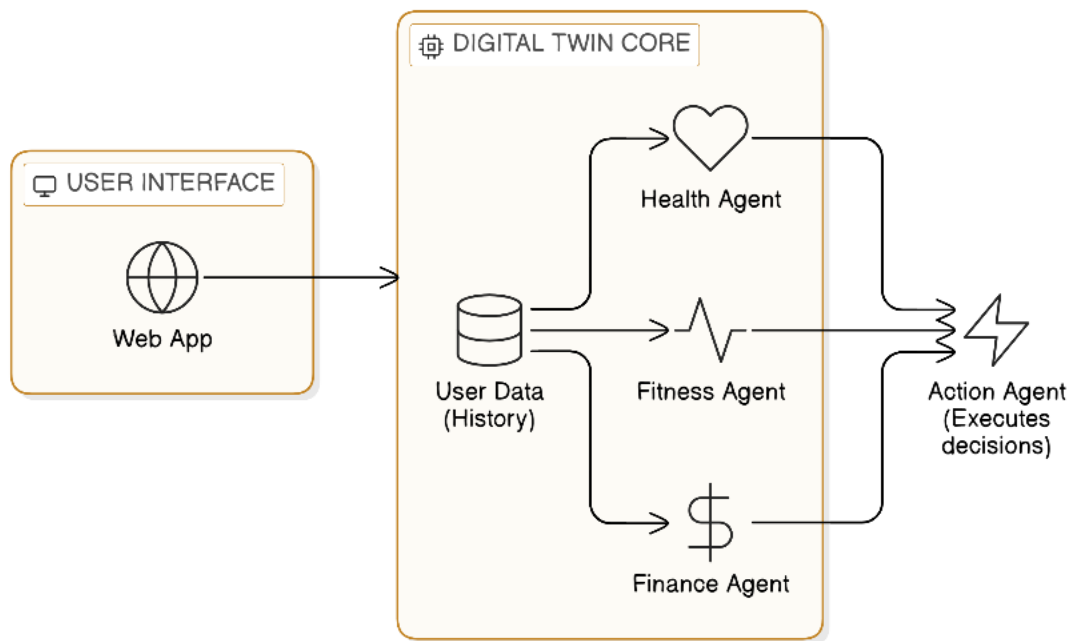


Figure 1 Agentic AI Architecture of the Personal Digital Twin System

7. System Functionality:

I. Health & Medical Intelligence Agent:

Purpose: Actively manages the user's medical and health data.

Key Features:

- Upload medical PDFs (lab results, reports, prescriptions).
- Summarize content in **plain language**.
- Track **trends over time**: vitals, blood tests, health markers.
- **Flag anomalies automatically** (e.g., high blood pressure, abnormal lab results).
- Generate **alerts and recommendations** for preventive action.
- Sends **proactive notifications** if trends indicate risk.
- Suggests **doctor visits or lifestyle changes** autonomously.
- Integrates with **fitness and nutrition agent** to recommend based on medical needs.
- **Explainability**: Shows justification for alerts

II. Fitness & Nutrition Agent:

Purpose: Acts as a personal coach, **planning and adjusting workouts and meals autonomously**.

Key Features:

- Suggest workouts aligned with user goals (weight loss, strength, endurance).
- Suggest meal plans based on dietary preferences and goals.
- Track progress and adapt plans **in real-time**.
- Provide reminders and motivation cues.
- Automatically updates weekly plans based on **progress or setbacks**.
- Alerts the user when workouts are missed or goals are not on track.
- Adjusts meal recommendations based on health data (e.g., blood sugar, BMI trends).

III. Finance & Savings Agent:

Purpose: Actively manages the user's personal finance and spending.

Key Features:

- Monitor bank transactions, categorize spending.
- Predict account balances and monthly savings potential.
- Suggest **autonomous financial actions**: reduce spending, move money to savings, alerts for upcoming bills.
- Sends notifications: "Overspending detected — consider limiting X this week."
- Suggests savings transfers automatically (user-approved).
- Tracks budgets and alerts on unusual spending patterns.

IV. Action & Automation Agent:

Purpose: Executes tasks and interventions actively on behalf of the user.

Key Features:

- Schedule gym sessions, study blocks, meetings.
- Send reminders and notifications proactively.

- Draft/send emails or messages in a user-adjustable style.
- Log all actions for transparency and review.
- **Calls external APIs and tools** (Google Calendar, Gmail/Outlook, Twilio...) to execute real-world actions autonomously.
- Adjusts actions dynamically if new events arise (e.g., skipped workout → reschedule automatically).

V. Digital Twin Goals Agent (Core Brain)

Purpose: Acts as the **central reasoning hub, prioritizing and coordinating actions** across all agents.

Key Features:

- Integrates outputs from Health, Fitness, and Finance agents.
- Decides what **matters most today**.
- Generates daily and weekly action plans.
- Suggests adjustments if multiple goals conflict.
- Provides **daily AI Life Report**: health summary, spending, productivity, and recommendations.

8. Technology Stack:

Layer	Technology Used
Frontend	React.js, Tailwind CSS
Agent Framework	Microsoft Agent Framework or LangChain
Backend / API	FastAPI (Python)
AI / ML	HuggingFace Transformers, OpenAI APIs
PDF / Document Parsing	pdfplumber, PyMuPDF
Database	SQL Server (structured data), MongoDB (unstructured data / reports)
Authentication	JWT, Role-Based Access Control (RBAC)
External Tool Integration	Google Calendar API, Gmail/Outlook API, Twilio
Microservices	Separate services for Frontend, Backend, and AI projects
Deployment	Docker, Docker Compose

9. Evaluation & Metrics

- **Health Agent:** Accuracy of medical report summaries, correct anomaly detection.
- **Fitness Agent:** Progress toward goals, adherence to recommended workouts/meals.
- **Finance Agent:** Accuracy of balance predictions, effectiveness of spending alerts.
- **Action Agent:** Success rate of automated tasks and scheduling.
- **System-Level:** User satisfaction and usefulness of daily AI Life Report.
- **Continuous Learning:** All agents improve over time by learning from user behavior and feedback, adapting recommendations automatically.